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2026 NCE Specimen Examination

# 2026

## MATHEMATICS

**Paper 1 of 4** | Number & Indices · Linear Equations · Pythagoras ·

Statistics

**Time Allowed:** 2 hours    **Total Marks:** 100    **Candidates:** Grade 9 NCE

Index Number: \_\_\_\_\_

### Formulae

**Area:** Rectangle  $A=lw$  | Triangle  $A=\frac{1}{2}bh$  | Circle  $A=\pi r^2$

**Pythagoras:**  $a^2+b^2=c^2$  (c=hypotenuse)

**Index laws:**  $a^m \times a^n = a^{m+n}$  |  $a^m \div a^n = a^{m-n}$  |  $(a^m)^n = a^{mn}$

**Volume:** Cuboid  $V=lwh$  | Cylinder  $V=\pi r^2 h$  | Cone  $V=\frac{1}{3}\pi r^2 h$

**Trig:**  $\sin\theta = \text{opp/hyp}$  |  $\cos\theta = \text{adj/hyp}$  |  $\tan\theta = \text{opp/adj}$

**Mean:**  $\Sigma fx \div \Sigma f$  | Standard form:  $a \times 10^n$ ,  $1 \leq a < 10$

### Instructions to Candidates

1. Write your Index Number in the space provided above.
2. Answer ALL questions in Sections A, B and C.
3. In Section A, tick ONE box only for each question.
4. Show all working clearly in Sections B and C.
5. Marks available are shown in brackets [ ].
6. You may use a calculator.  $\pi=3.142$  unless otherwise stated.
7. This paper contains 28 questions totalling 100 marks.

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**SECTION A — Multiple Choice Questions (10 marks) [Q1–10, 1 mark each]**

For each question tick ONE box only.

- 1** Simplify  $2^3 \times 2^5$ . [1]
- ☐ A) 2 8
- ☐ B) 4 8
- ☐ C) 2 15
- ☐ D) 4 15
- 2** Simplify  $(5^2)^3$ . [1]
- ☐ A) 5 5
- ☐ B) 5 6
- ☐ C) 25 6
- ☐ D) 5 9
- 3** The value of  $6^0$  is: [1]
- ☐ A) 0
- ☐ B) 6
- ☐ C) 1
- ☐ D) undefined
- 4** Solve  $4x-3=17$ . [1]
- ☐ A)  $x=3.5$
- ☐ B)  $x=5$
- ☐ C)  $x=7$
- ☐ D)  $x=14$
- 5** A right-angled triangle has legs 6 cm and 8 cm. The hypotenuse is: [1]
- ☐ A)  $\sqrt{48}$  cm
- ☐ B) 10 cm
- ☐ C) 14 cm
- ☐ D) 100 cm
- 6** The mean of 4, 7, 9, 12, 13 is: [1]
- ☐ A) 7

- ☐ B) 9
- ☐ C) 8.5
- ☐ D) 45

7 The value of  $3^{-3}$  is:

[1]

- ☐ A) -27
- ☐ B)  $1/9$
- ☐ C)  $1/27$
- ☐ D)  $-1/27$

8 Solve  $3(2x+1)=21$ .

[1]

- ☐ A)  $x=3$
- ☐ B)  $x=4$
- ☐ C)  $x=3.5$
- ☐ D)  $x=10/3$

9 A right-angled triangle has hypotenuse 13 cm and one leg 5 cm. The other leg is:

[1]

- ☐ A) 8 cm
- ☐ B) 12 cm
- ☐ C)  $\sqrt{194}$  cm
- ☐ D) 144 cm

10 The median of  $\{3, 7, 9, 14, 20\}$  is:

[1]

- ☐ A) 7
- ☐ B) 9
- ☐ C) 11.5
- ☐ D) 13

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## SECTION B — Short Structured Questions (48 marks) [Q11–22]

Show all working. Answers without working may not receive full marks.

### 11 Laws of Indices

[3]

(a) Simplify  $7^2 \times 7^3$ .

[1]

(b) Simplify  $(4^3)^2$ .

[1]

(c) Write  $8^{-1}$  as a fraction.

[1]

### 12 Standard Form

[3]

(a) Write 47 000 in standard form.

[1]

(b) Write  $2.8 \times 10^{-2}$  as an ordinary number.

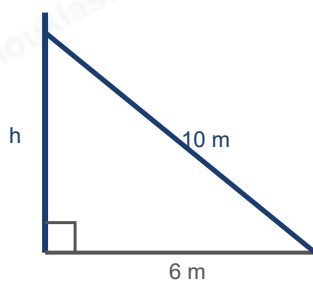
[1]

(c) Evaluate  $5^0 + 2^{-1}$ . Give your answer as a fraction.

[1]

**13 Linear Equations****[4]****(a)** Solve  $5x+8=33$ .**[2]****(b)** Solve  $4x-3=2x+9$ .**[2]****14 Pythagoras — Ladder Problem****[4]**

A 10 m ladder rests against a vertical wall with its foot 6 m from the base.

**(a)** Write down Pythagoras' theorem for this triangle.**[1]****(b)** Calculate the height  $h$ .**[3]**

**15 Statistics — Mean from Frequency Table****[4]**

The marks scored by 25 students are shown below.

| Mark (x) | 2 | 3 | 4 | 5 | 6 |
|----------|---|---|---|---|---|
| Freq (f) | 3 | 5 | 8 | 6 | 3 |
| fx       |   |   |   |   |   |

(a) Complete the fx row.

**[2]**

(b) Calculate the mean mark.

**[2]****16 Fractional and Negative Indices****[4]**

(a) Evaluate  $25^{1/2}$ .

**[1]**

(b) Evaluate  $64^{1/3}$ .

**[1]**

(c) Evaluate  $16^{3/4}$ . Show your steps.

**[2]**

**17 Linear Equations — Word Problem**

[4]

Sam thinks of a number. He multiplies it by 5 and subtracts 8. The result is 32.

(a) Let  $n$  be Sam's number. Form an equation.

[1]

(b) Solve for  $n$ .

[2]

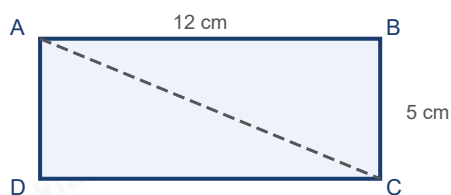
(c) Verify your answer.

[1]

**18 Pythagoras — Rectangle Diagonal**

[5]

Rectangle ABCD has  $AB=12$  cm and  $BC=5$  cm.



(a) Calculate the length of diagonal AC.

[3]



(b) What type of triangle is ABC? Give a reason. [1]

(c) Calculate the perimeter of triangle ABC. [1]

**19 Statistics — Mode, Median, Range**

[4]

Data: 5, 8, 3, 9, 5, 12, 7, 5, 10, 8

(a) Write the data in ascending order. [1]

(b) Find the median. [1]

(c) Find the mode. [1]

(d) Find the range. [1]

**20 Index Laws with Algebra**

[4]

(a) Simplify  $a^5 \times a^{-3}$ .

[1]

(b) Simplify  $(2x^3)^4$ .

[2]

(c) Simplify  $b^8 \div b^5$ .

[1]

**21 Solving Exponential Equations**

[5]

(a) Given  $2^n = 64$ , find  $n$ .

[1]

(b) Write down the value of  $2^{n+1}$ .

[1]

(c) Write down the value of  $2^{n-2}$ .

[1]

(d) Solve  $2^x = 1/32$ . [2]



**22 Statistics — Mean and Median**

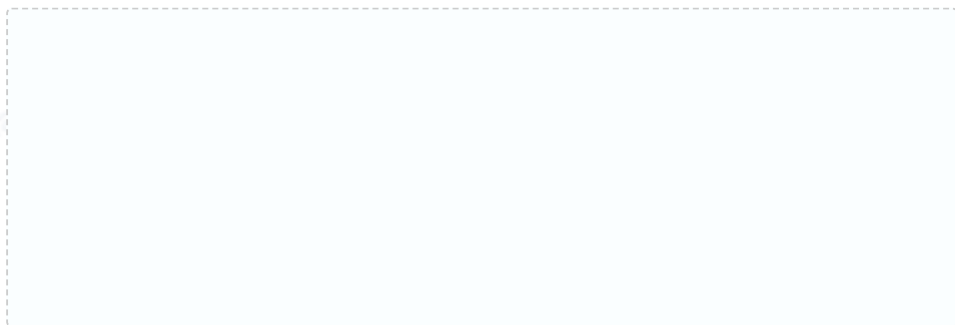
[4]

Ages of 8 players: 15, 17, 16, 18, 15, 19, 16, 20

(a) Find the mean age. [2]



(b) Find the median age. Show your working. [2]



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**SECTION C — Long Structured Questions (42 marks) [Q23–28, 7 marks each]**

Show ALL working. Marks may be awarded for correct method even if the final answer is wrong.

**23 Indices — Comprehensive****[7]**

- (a) Evaluate  $4^{3/2}$ . Show all steps.

**[2]**

- (b) Simplify  $(x^2y^3)^4 \div (x^3y^2)^2$ .

**[3]**

- (c) Given  $3^x=81$ , find  $x$ .

**[1]**

- (d) Evaluate  $5^{-2} \times 25$ .

**[1]**

**24 Linear Equations — Extended****[7]**

- (a) Solve  $3(x-4)=2(x+1)$ . Show all steps.

**[3]**

- (b) The sum of three consecutive integers is 78. Let the smallest be  $n$ .

- (i) Form an equation in  $n$ .

**[1]**

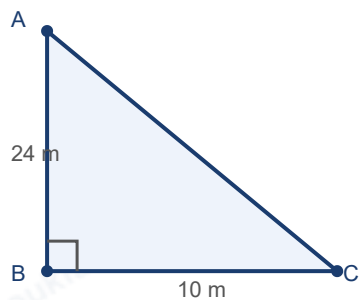
- (ii) Solve to find the three integers.

**[2]**

- (iii) Verify your answer.

**[1]****25 Pythagoras — Extended Problem****[7]**

A triangular field has a right angle at B, with  $AB=24$  m and  $BC=10$  m.



- (a) Calculate the length of AC. [3]

- (b) A fence post D lies on AC such that BD is perpendicular to AC. Using the area of triangle ABC, calculate the length BD. Show all working. [4]

## 26 Statistics — Grouped Frequency

The marks of 40 students are given below.

[7]

| Marks     | 20–30 | 30–40 | 40–50 | 50–60 | 60–70 |
|-----------|-------|-------|-------|-------|-------|
| Frequency | 5     | 8     | 12    | 10    | 5     |

- (a) Write down the modal class. [1]

- (b) Using midpoints 25, 35, 45, 55, 65, calculate an estimate of the mean mark. Show all working. [4]

- (c) State which class the median lies in and justify with cumulative frequencies. [2]

**27 Pythagoras — Rectangular Field** [7]  
Rectangular field PQRS has  $PQ=40$  m and  $QR=30$  m.

- (a) Calculate the diagonal PR. [2]

- (b) M is the midpoint of QR. Calculate PM, correct to 2 decimal places. [3]

(c) Calculate the area of triangle PRS. [2]

**28 Statistics — Data Analysis**

[7]

A survey of 30 students recorded how many books each read last month.

| Books (x) | 0 | 1 | 2 | 3 | 4 | 5 |
|-----------|---|---|---|---|---|---|
| Freq (f)  | 2 | 5 | 8 | 9 | 4 | 2 |

(a) Calculate the mean number of books read. [3]

(b) Find the mode and the median. [2]

- (c) The teacher claims the average student reads 3 books per month. Which measure of average supports this? Explain. [2]

# Mark Scheme — Paper 1 (Number & Indices · Linear Equations · Pythagoras · Statistics) 2026

Equivalent correct methods receive full credit. M=method mark; A=accuracy mark. Allow ft for consistent errors.

| Q                | Answer / Working                              | Marks | Notes                             |
|------------------|-----------------------------------------------|-------|-----------------------------------|
| <b>SECTION A</b> |                                               |       |                                   |
| 1                | A — $2^8$ (add indices: $3+5=8$ )             | 1     |                                   |
| 2                | B — $5^6$ (multiply indices: $2 \times 3=6$ ) | 1     |                                   |
| 3                | C — 1                                         | 1     |                                   |
| 4                | B — $x=5$ ( $4x=20$ )                         | 1     |                                   |
| 5                | B — 10 cm ( $\sqrt{36+64}=10$ )               | 1     |                                   |
| 6                | B — 9 (sum=45, mean=9)                        | 1     |                                   |
| 7                | C — $1/27$                                    | 1     |                                   |
| 8                | A — $x=3$ ( $6x=18$ )                         | 1     |                                   |
| 9                | B — 12 cm ( $\sqrt{169-25}=12$ )              | 1     |                                   |
| 10               | B — 9                                         | 1     |                                   |
| <b>SECTION B</b> |                                               |       |                                   |
| 11(a)            | $7^5$                                         | 1     |                                   |
| 11(b)            | $4^6$ (=4096)                                 | 1     |                                   |
| 11(c)            | $1/8$                                         | 1     |                                   |
| 12(a)            | $4.7 \times 10^4$                             | 1     |                                   |
| 12(b)            | 0.028                                         | 1     |                                   |
| 12(c)            | $1 + 1/2 = 3/2$                               | 1     |                                   |
| 13(a)            | $5x=25$ ; <b><math>x=5</math></b>             | 2     | M1 collecting;<br>A1              |
| 13(b)            | $2x=12$ ; <b><math>x=6</math></b>             | 2     | M1 collecting;<br>A1              |
| 14(a)            | $h^2+6^2=10^2$ (or equivalent)                | 1     |                                   |
| 14(b)            | $h^2=100-36=64$ ; <b><math>h=8</math> m</b>   | 3     | M1 Pythagoras;<br>M1 subtract; A1 |
| 15(a)            | fx: 6, 15, 32, 30, 18 ( $\Sigma fx=101$ )     | 2     | 1 for any 3 correct               |

|                  |                                                               |   |                                     |
|------------------|---------------------------------------------------------------|---|-------------------------------------|
| 15(b)            | $101 \div 25 = 4.04$                                          | 2 | M1; A1                              |
| 16(a)            | <b>5</b>                                                      | 1 |                                     |
| 16(b)            | <b>4</b>                                                      | 1 |                                     |
| 16(c)            | $16^{1/4}=2$ ; $2^3=8$                                        | 2 | M1; A1                              |
| 17(a)            | $5n-8=32$                                                     | 1 |                                     |
| 17(b)            | $5n=40$ ; <b><math>n=8</math></b>                             | 2 | M1; A1                              |
| 17(c)            | $5(8)-8=32$ ✓                                                 | 1 |                                     |
| 18(a)            | $AC^2=144+25=169$ ; <b><math>AC=13</math> cm</b>              | 3 | M1 Pythagoras;<br>M1 compute; A1    |
| 18(b)            | Right-angled ( $5^2+12^2=13^2$ confirms Pythagoras)           | 1 |                                     |
| 18(c)            | $5+12+13=30$ cm                                               | 1 |                                     |
| 19(a)            | 3, 5, 5, 5, 7, 8, 8, 9, 10, 12                                | 1 |                                     |
| 19(b)            | $(7+8)/2=7.5$                                                 | 1 |                                     |
| 19(c)            | <b>5</b> (appears 3 times)                                    | 1 |                                     |
| 19(d)            | $12-3=9$                                                      | 1 |                                     |
| 20(a)            | $a^2$                                                         | 1 |                                     |
| 20(b)            | <b><math>16x^{12}</math></b>                                  | 2 | M1; A1                              |
| 20(c)            | $b^3$                                                         | 1 |                                     |
| 21(a)            | <b><math>n=6</math></b>                                       | 1 |                                     |
| 21(b)            | $2^7=128$                                                     | 1 |                                     |
| 21(c)            | $2^4=16$                                                      | 1 |                                     |
| 21(d)            | $1/32=2^{-5}$ ; <b><math>x=-5</math></b>                      | 2 | M1; A1                              |
| 22(a)            | sum=136; mean= <b>17</b>                                      | 2 | M1; A1                              |
| 22(b)            | Ordered: 15,15,16,16,17,18,19,20;<br>median= $(16+17)/2=16.5$ | 2 | M1<br>order+select; A1              |
| <b>SECTION C</b> |                                                               |   |                                     |
| 23(a)            | $\sqrt{4}=2$ ; $2^3=8$                                        | 2 | M1; A1                              |
| 23(b)            | $x^8y^{12} \div x^6y^4 = x^2y^8$                              | 3 | M1 expand<br>each; M1 divide;<br>A1 |
| 23(c)            | $3^4=81$ ; <b><math>x=4</math></b>                            | 1 |                                     |
| 23(d)            | $(1/25) \times 25 = 1$                                        | 1 |                                     |
| 24(a)            | $3x-12=2x+2$ ; <b><math>x=14</math></b>                       | 3 | M1 expand; M1<br>collect; A1        |
| 24(b)<br>(i)     | $n+(n+1)+(n+2)=78 \rightarrow 3n+3=78$                        | 1 |                                     |

|                |                                                                                                                                               |            |                                                  |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------------------------------------------|
| 24(b)<br>(ii)  | $3n=75$ ; $n=25$ ; integers: <b>25, 26, 27</b>                                                                                                | 2          | M1; A1 all three                                 |
| 24(b)<br>(iii) | $25+26+27=78$ ✓                                                                                                                               | 1          |                                                  |
| 25(a)          | $AC^2=576+100=676$ ; <b>AC=26 m</b>                                                                                                           | 3          | M1 Pythagoras;<br>M1 values; A1                  |
| 25(b)          | Area= $\frac{1}{2} \times 24 \times 10=120$ ; also= $\frac{1}{2} \times 26 \times BD$ ;<br>$BD=240/26=\mathbf{120/13 \approx 9.23 \text{ m}}$ | 4          | M1 area; M1<br>second formula;<br>M1 algebra; A1 |
| 26(a)          | Modal class: <b>40–50</b>                                                                                                                     | 1          |                                                  |
| 26(b)          | fx: $125+280+540+550+325=1820$ ;<br>mean= $1820/40=\mathbf{45.5}$                                                                             | 4          | M1 midpoints;<br>M1 fx; M1 $\Sigma$ ; A1         |
| 26(c)          | Cumulative: 5, 13, 25, ... 20th value in <b>40–50</b>                                                                                         | 2          | M1 cumulative;<br>A1                             |
| 27(a)          | $PR^2=1600+900=2500$ ; <b>PR=50 m</b>                                                                                                         | 2          | M1; A1                                           |
| 27(b)          | QM=15; $PM^2=1600+225=1825$ ;<br>PM= <b>42.72 m</b>                                                                                           | 3          | M1 QM=15; M1<br>Pythagoras; A1                   |
| 27(c)          | $\frac{1}{2} \times \text{area of rectangle} = \frac{1}{2} \times 1200 = \mathbf{600 \text{ m}^2}$                                            | 2          | M1; A1                                           |
| 28(a)          | $\Sigma fx = 0+5+16+27+16+10=74$ ;<br>mean= $74/30 \approx \mathbf{2.47}$                                                                     | 3          | M1 fx; M1 sum;<br>A1                             |
| 28(b)          | Mode= <b>3</b> ; Median: 15th+16th both = <b>3</b>                                                                                            | 2          | M1 each; A1                                      |
| 28(c)          | <b>Mode and median</b> both equal 3; the<br>mean (2.47) does not support the claim.                                                           | 2          | M1 identify<br>measure; A1<br>explanation        |
| <b>TOTAL</b>   |                                                                                                                                               | <b>100</b> |                                                  |